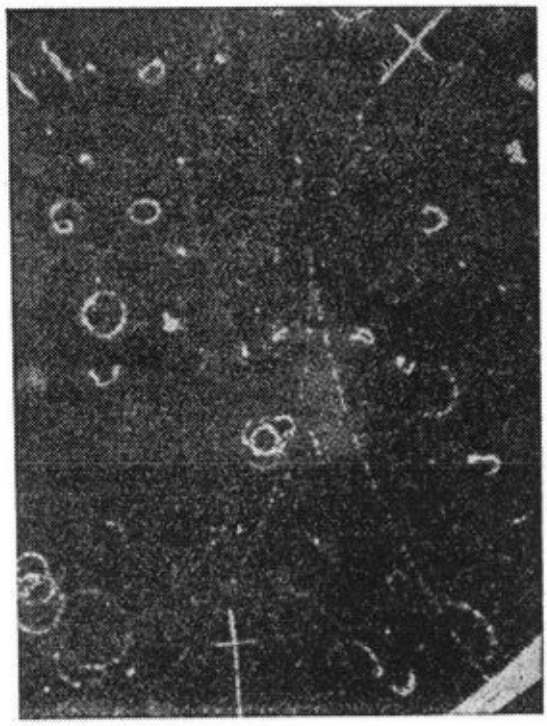


Figure 9.—Pair production in the cloud chamber

However, this phenomenon of pair formation must not be regarded as indicating that a photon is actually composed of an electron and a positron. A photon is a true elementary particle, in the strictest sense of the word. But it is capable of changing, when it enters into interaction with other particles or with powerful fields. Generally speaking, the concept of ‘elementary particles’ in modern physics has undergone a change; these elementary particles may be described as the ‘ultimate, indivisible building blocks of matter’ in a very limited sense only. For it has been proved that these elementary particles can change into each other practically without restriction, so long as it is compatible with the laws of conservation of mass, energy, etc. But just for this very reason, it is meaningless to describe any of them as being composed of some of the others.

Beta particles have a far longer range than alpha rays. The explanation is not that the former have more energy, but principally that due to their smaller charge and greater velocity, they have a considerably smaller capacity to produce ionization, and therefore they lose their energy much more slowly along their path.

But there is another very characteristic difference between alpha and beta rays. All alpha rays of a homogeneous radioactive substance have exactly the same range, and therefore, the same energy. This is to be expected, for as the energy released in any chemical reaction is determined both by the initial and end states of the system, so must also the energy released by radioactive decay—in other words, essentially, the energy of the alpha particle—depend